Sparse Non-Negative Activations: Why Banning Negative Numbers Changes What a Network Learns

One-page concept summary. DataForge 2026, Pathway Track. Topic: Sparse Non-Negative Activations.

Dense Transformers activate nearly every unit on every forward pass. GELU and SiLU, the activation functions behind most modern LLMs, allow both positive and negative outputs, so a network can represent a feature by combining pushes in opposite directions. That flexibility comes at a cost: individual units end up encoding several unrelated concepts at once, a property researchers call polysemanticity, which makes networks harder to interpret and harder to prune without also slowing them down.

Sparse non-negative activations remove one specific freedom. A unit's activation is clamped to zero whenever it would go negative, most simply through a ReLU. Two consequences follow directly from that single constraint. First, subtraction becomes impossible during accumulation: once a unit's output is added into a later layer, nothing downstream can cancel it, so the network has to represent a concept as a sum of strictly positive, largely non-overlapping parts rather than a difference of opposing pushes. Second, whichever units land below zero contribute exactly nothing to the forward pass, not a small amount but zero, so the fraction of a layer doing any work at all becomes a directly observable, variable quantity instead of a fixed hyperparameter.

That second point is what this concept turns on: how many units fire is not set by a quota, it is set by how predictable the input is. A familiar pattern reuses a small, well worn set of units; an unfamiliar one recruits more of the network. This is the mechanism Pathway's Dragon Hatchling (BDH) architecture builds its entire design around. BDH couples ReLU style non-negative activations with local, Hebbian synaptic updates, and its authors report that roughly 5 percent of neurons are active for a given input in trained runs, with that fraction varying with how predictable the input is rather than being fixed by design (Kosowski et al., 2025). That figure is developer reported, from Pathway's own paper, not an independent third party benchmark of BDH itself.

Independent academic work supports the underlying mechanism, if not BDH's specific number. Google Research's Lazy Neuron Phenomenon study measured naturally emerging activation sparsity in ordinary, already trained ReLU Transformers with no connection to BDH, finding as few as 3.0 percent of entries nonzero per input in T5-Base, with larger models showing even more sparsity (Li et al., 2022, ICLR 2023). Apple's ReLU Strikes Back goes further and argues for deliberately reinstating ReLU in modern LLMs specifically to reclaim this property, showing it cuts inference computation with a negligible accuracy cost (Mirzadeh et al., 2023, ICLR 2024).

BDH-CQ, Pathway's later reasoning system, has no direct role here. Its contribution is a recurrent latent workspace for in-context learning on tasks like ARC-AGI, not the activation level sparsity mechanism this concept is about (Pathway Research, 2026). Folding it in would suggest a connection this topic does not actually support.

The clearest open gap is independent verification. No outside team has yet reproduced BDH's specific 5 percent figure, or tested its claim that sparsity tracks predictability rather than a fixed quota, on BDH itself. The Lazy Neuron and ReLU Strikes Back results are consistent with the mechanism being real and general, but neither was run on BDH, and a benchmark score from a different model is not the same evidence as an external reproduction of a specific reported number.

System	Sparsity's role	Interpretability link	Evidence type
Dense Transformer (GELU/SiLU)	Absent by design	Units are polysemantic	Not applicable, no sparsity is claimed
ReLU Transformer (efficiency work)	Exploitable side effect, not a design goal	Not a stated goal	Independent, peer reviewed, reproducible
BDH (Dragon Hatchling)	Organizing design principle	Monosemantic synapses reported	Developer reported, not yet independently reproduced

Primary sources

Kosowski, A., Uznanski, P., Chorowski, J., Stamirowska, Z., and Bartoszkiewicz, M. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. arXiv:2509.26507.

Li, Z., You, C., Bhojanapalli, S., Li, D., Rawat, A. S., Reddi, S. J., Ye, K., Chern, F., Yu, F., Guo, R., and Kumar, S. (2022). The Lazy Neuron Phenomenon: On Emergence of Activation Sparsity in Transformers. arXiv:2210.06313 (ICLR 2023).

Mirzadeh, I., Alizadeh-Vahid, K., Mehta, S., Del Mundo, C. C., Tuzel, O., Samei, G., Rastegari, M., and Farajtabar, M. (2023). ReLU Strikes Back: Exploiting Activation Sparsity in Large Language Models. arXiv:2310.04564 (ICLR 2024).

Pathway Research (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. arXiv:2608.09888. Referenced only to state that it is not load-bearing evidence for this topic.